

Credit Card Approval Prediction – Sprint Planning

Sprint

A **Sprint** is a fixed period during which the development team completes a set of planned tasks. For the Credit Card Approval Prediction project, the work is divided into five sprints covering data collection, analysis, preprocessing, model development, and web application deployment.

Epic

An **Epic** is a major feature or functionality that is too large to complete in a single sprint. Each epic is divided into smaller user stories that can be completed within a sprint.

Story

A **User Story** is a small unit of work representing a specific functionality of the system.

Story Points

Story points estimate the effort required to complete each user story.

Story Point	Difficulty
1	Very Easy
2	Easy
3	Moderate
5	Difficult

Sprint 1 – Project Planning & Data Collection

Epic 1: Project Setup

User Story	Story Points
USN-1 Create GitHub Repository	2
USN-2 Prepare Project Workflow	2
USN-3 Create ER Diagram	3

Epic 2: Data Collection

User Story	Story Points
USN-4 Download Dataset	2
USN-5 Import Required Libraries	1
USN-6 Read Dataset	2

Total Story Points = 12

Sprint 2 – Data Analysis

Epic 3: Exploratory Data Analysis

User Story	Story Points
USN-7 Univariate Analysis	2
USN-8 Multivariate Analysis	3
USN-9 Descriptive Statistics	2
USN-10 Data Visualization	3
USN-11 Correlation Analysis	2

Total Story Points = 12

Sprint 3 – Data Preprocessing

Epic 4: Data Preparation

User Story	Story Points
USN-12 Remove Duplicate Records	2
USN-13 Handle Missing Values	3
USN-14 Data Cleaning	2
USN-15 Feature Engineering	3
USN-16 Encode Categorical Data	2

Total Story Points = 12

Sprint 4 – Machine Learning Model

Epic 5: Model Development

User Story	Story Points
USN-17 Logistic Regression	3
USN-18 Random Forest	3
USN-19 Decision Tree	3
USN-20 Compare Model Accuracy	3

Total Story Points = 12

Sprint 5 – Web Application & Deployment

Epic 6: Application Development

User Story	Story Points
USN-21 Design HTML Pages	2
USN-22 Develop Flask Application	3
USN-23 Test the Application	2
USN-24 Deploy Application	3
USN-25 Final Documentation & Conclusion	2

Total Story Points = 12

Total Story Points

Sprint	Story Points
Sprint 1	12
Sprint 2	12
Sprint 3	12
Sprint 4	12
Sprint 5	12

Total Story Points = 60

Velocity Calculation

Formula

$$\begin{aligned}\text{Velocity} &= \frac{\text{Total Story Points Completed}}{\text{Number of Sprints}} \\ &= \frac{60}{5} = 12\end{aligned}$$

Team Velocity

12 Story Points per Sprint